

LYMPHOID NEOPLASIA

Expression of cFLIP in B cells is essential for diffuse large B-cell lymphoma pathogenesis

Kristie T. Bariboloka,¹ Santiago Serrano-Saenz,² Deniz Pinar Savcigil,¹ Sarah Spöck,³ Rebecca E. Poss,^{4,5} Luca D. Schreurs,^{1,2,6} Jens Löber,⁷ Nazli Serin,⁷ Justinas Valiulis,^{1,8} Kalvin Nugraha,⁹ Giulia Gangarossa,¹⁰ Anna Lütz,^{1,2,6,11,12} Moritz Reese,^{1,2,6,12} Sebastian Klein,¹³ Julius C Enssle,^{4,5,14} Reinhard Buettner,¹⁵ Hamid Kashkar,^{1,2,16} Martin Peifer,^{1,7} Björn Chapuy,^{6,17} Gero Knittel,¹³ Verena Labi,³ Andreas Villunger,^{3,18} Phuong-Hien Nguyen,^{1,2,6} Sebastian Scheich,^{4,5,14} Thomas Oellerich,^{4,5,14,19,20} Henning Walczak,^{2,21,22} Ruth Flümann,^{1,2,6} Ron D. Jachimowicz,^{1,2,6,12} Michael Hallek,^{1,2,6} Hans Christian Reinhardt,¹³ and Alessandro Annibaldi¹

¹Center for Molecular Medicine Cologne and ²Cologne Excellence Cluster on Cellular Stress Responses in Aging-Associated Diseases, University of Cologne, Cologne, Germany; ³Institute for Developmental Immunology, Biocenter, Medical University of Innsbruck, Innsbruck, Austria; ⁴Department of Hematology and Oncology, Goethe University Frankfurt, University Hospital Frankfurt am Main, Frankfurt, Germany; ⁵Frankfurt Cancer Institute, Goethe University, Frankfurt, Germany; ⁶Department I of Internal Medicine, Center for Integrated Oncology Aachen Bonn Cologne Düsseldorf, Faculty of Medicine and University Hospital Cologne, University of Cologne, Cologne, Germany; ⁷Department of Hematology, Oncology, and Cancer Immunology, Campus Benjamin Franklin, Charité-Universitätsmedizin Berlin, corporate member of Freie Universität Berlin and Humboldt-Universität zu Berlin, Berlin, Germany; ⁸Department of Translational Genomics, Faculty of Medicine and University Hospital Cologne, University of Cologne, Cologne, Germany; ⁹Department of Immunobiology, University of Lausanne, Epalinges, Switzerland; ¹⁰Department of Experimental and Clinical Biomedical Sciences, University of Florence, Florence, Italy; ¹¹Mildred Scheel School of Oncology Aachen Bonn Cologne Düsseldorf, Faculty of Medicine and University Hospital of Cologne, Cologne, Germany; ¹²Max Planck Institute for Biology of Ageing, Cologne, Germany; ¹³Department of Haematology and Stem Cell Transplantation, West German Cancer Center, German Cancer Consortium Partner Site Essen, Center for Molecular Biotechnology, University Hospital Essen, University of Duisburg-Essen, Essen, Germany; ¹⁴German Cancer Consortium Partner Site Frankfurt/Mainz, Frankfurt, Germany; ¹⁵Institute of Pathology and ¹⁶Institute for Molecular Immunology, Faculty of Medicine and University Hospital of Cologne, University of Cologne, Cologne, Germany; ¹⁷German Cancer Consortium, partner site Berlin, a partnership between German Cancer Consortium and Charité-Universitätsmedizin Berlin, Berlin, Germany; ¹⁸The Research Center for Molecular Medicine of the Austrian Academy of Sciences, Vienna, Austria; ¹⁹German Cancer Research Center, Heidelberg, Germany; ²⁰University Cancer Center, Goethe University, Frankfurt, Germany; ²¹Center for Biochemistry, Institute of Biochemistry I, University of Cologne, Cologne, Germany; and ²²UCL Cancer Institute, University College London, London, United Kingdom

KEY POINTS

- cFLIP-mediated inhibition of extrinsic apoptosis cooperates with BCL-2-dependent intrinsic apoptosis blockade to support DLBCL development.
- Reactivation of extrinsic apoptosis by CFLAR deletion efficiently kills ABC-DLBCL cells, in spite of BCL2 overexpression.

Diffuse large B-cell lymphoma (DLBCL) is a highly heterogeneous malignant disease that remains a major clinical challenge, as relapsed and refractory disease is difficult to treat. Apoptosis evasion is a major feature of DLBCL. However, while the suppression of intrinsic apoptosis has long been recognized as a lymphoma-promoting event, the role of extrinsic apoptosis has remained poorly defined. In this study, we demonstrated at the genetic level that expression of cellular Fas-associated death domain protein-like IL-1 β -converting enzyme-inhibitory protein (cFLIP), the most crucial, non-redundant inhibitor of extrinsic apoptosis, in B cells is necessary for the development of DLBCL in an autochthonous murine model. Indeed, B-cell-specific deletion of *Cflar*, the gene encoding for cFLIP, prevented lymphomagenesis mediated by oncogenic *Myd88* and overexpression of *BCL2*. In human lymphoma cells, we showed that the absence of cFLIP sensitized activated B-cell-like (ABC)- but not germinal center B-cell-like (GCB)-DLBCL subtype cells to TRAIL- or lipopolysaccharide-induced, caspase-8-mediated apoptosis. Furthermore, we unveiled a cell death-independent role of cFLIP in the suppression of proinflammatory cytokines at the transcriptional level, selectively in the ABC subtype. These results indicate that the suppression of intrinsic apoptosis can support lymphomagenesis only if extrinsic apoptosis is properly controlled. Moreover, licensing extrinsic apoptosis through *CFLAR* deletion can efficiently promote death in DLBCL cells, despite the suppression of the intrinsic pathway. Overall, these data provide a rationale for the development of cFLIP inhibitors for the treatment of ABC-DLBCL and possibly other hematological cancers.

Introduction

Diffuse large B-cell lymphoma (DLBCL) represents ~35% of all cases of non-Hodgkin lymphomas (NHL). It is the most common aggressive lymphoma, and it is a clinically and biologically highly heterogeneous disease.¹⁻³ Although frontline chemotherapy (R-CHOP [rituximab, cyclophosphamide, doxorubicin, oncovin (vincristine), and prednisone] like) achieves cure rates of ~60%, refractory and/or relapsed disease remains a major clinical challenge.⁴ Therefore, there is a need to find novel vulnerabilities for the treatment of DLBCL.

DLBCL is transcriptionally subdivided into activated B-cell-like (ABC)- and germinal center (GC) B-cell-like (GCB)-DLBCL.² Recently, 2 comprehensive genomic analyses classified DLBCL into at least 5 genetically defined subtypes (DLBclass, LymphGen).⁵⁻⁹ DLBclass classifies all cases into 5 distinct clusters (cluster 1 [C1]-C5) with specific genetic signatures.^{5,7,8} C5 DLBCL (overlapping with the MCD DLBCL of LymphGen) comprises ABC-DLBCLs with substantially worse overall survival compared with the other tumors and is characterized by frequent *MYD88* and *CD79B* mutations and almost uniform copy number gain of the antiapoptotic oncogene *BCL2*.⁵ In addition to defects in the intrinsic apoptotic pathway, of which BCL-2 is a major inhibitor,¹⁰ DLBCLs can display dysregulation of the extrinsic apoptotic pathway^{11,12} that is mediated by a class of surface receptors called death receptors (DRs), which include tumor necrosis factor R1 (TNFR1), TRAIL-R1/R2, and FS-7-associated surface antigen (CD95).¹³ DR pathways are critical for B-cell homeostasis, and their deregulation has been associated with immune disorders, including lymphadenopathy, splenomegaly, and increased risk of B-cell lymphoma and systemic autoimmune disease.^{14,15} However, the exact role of extrinsic apoptosis suppression and whether this is redundant to the suppression of the intrinsic pathway in lymphoma development has never been addressed.

DRs, upon binding to their cognate ligands, transduce cell death-inducing signals that require the activation of the initiator caspase-8 and the executioner caspase-3 and 7, ultimately resulting in apoptosis.^{13,16-19} In some cell types, defined as type I cells, DR-induced caspase-8 activation is sufficient to activate the executioner caspases and effectively induce apoptosis. On the contrary, type II cells require the caspase-8-mediated engagement of the intrinsic apoptotic pathway for the activation of the executioner caspases.²⁰ An essential inhibitor of caspase-8-mediated apoptosis is cellular Fas-associated death domain protein-like-IL-1 β -converting enzyme-inhibitory protein (cFLIP), a close relative of caspase-8 that lacks catalytic activity. cFLIP, encoded by the gene *CFLAR*, binds with high affinity to caspase-8 and prevents its full, cleavage-mediated activation, thereby limiting its ability to induce apoptosis.²¹⁻²³ B-cell-specific *Cflar* deletion does not cause a significant phenotypic alteration, although primary B cells lacking cFLIP exhibit abnormally increased sensitivity to CD95L-induced cell death.^{24,25}

Consistent with its role in inhibiting apoptosis in B cells, elevated expression of cFLIP has been associated with different types of NHL, including DLBCL. High levels of cFLIP in DLBCL correlate with a lack of caspase-8 activation, disease progression, and poor outcomes for patients.^{26,27} Despite this evidence, the role of cFLIP in DLBCL pathogenesis and biology was never addressed. In this

study, we show that cFLIP is upregulated in DLBCL at both the transcriptional and protein levels, particularly in the ABC subtype. Moreover, B-cell-specific deletion of *Cflar* prevented lymphoma development in an autochthonous mouse model that is reminiscent of ABC C5/MCD DLBCL, driven by *Myd88*^{L252P} mutation (the murine orthologue of human *MYD88*^{L265P}) and *BCL2* overexpression.²⁸⁻³⁰ Furthermore, *CFLAR* deletion sensitizes human ABC- but not GCB-DLBCL cells to TRAIL- and LPS-induced caspase-8-mediated apoptosis and promotes transcriptional activation of proinflammatory genes selectively in ABC-DLBCL cells.

Taken together, our results demonstrate that cFLIP, one of the most essential inhibitors of apoptosis, represents a vulnerability in DLBCL; therefore, it is a promising therapeutic target.

Materials and methods

Experimental mice

The generation of the *CD19*^{Cre}, *Myd88*^{cond.p.L252P}, *ROSA26*^{LSL.BCL2-IRES-GFP}, and *Cflip*^F alleles has been described previously.^{28,31} For survival analyses, animals were recorded as events if they succumbed to disease or had to be euthanized upon reaching termination criteria.

All experimental procedures were approved by local authorities (Landesamt für Natur, Umwelt und Verbraucherschutz Nordrhein-Westfalen).

CFLAR transcript abundance in human cancers

For the primary human data sets, the publicly available gene expression data sets GSE12453,³² GSE98588,⁵ and GSE181063 were assessed through Gene Expression Omnibus. RNA-sequencing (RNA-seq) data⁷ and cell line data^{33,34} were also downloaded.

RNA-seq

RNA was isolated from cells (5×10^6 cells) using a commercial kit (R1055; Zymo Research) in triplicate. Sequencing reads were mapped to the human reference genome (GRCh38) using STAR. Transcript quantification was performed using RSEM. The acquired transcriptomic profiles were then compared across different cell lines and their genotypes by applying differential expression analysis based on the negative binomial distribution. The differential expression model was built in a way that it measures the effect brought by the knock-out genotype for both cell lines separately. The resulting list of genes was ranked based on the stat parameter and used for gene set enrichment analysis. The featured gene sets were retrieved from the Molecular Signatures Database.

Cell lines

DLBCL cell lines were provided by the Chapuy laboratory (Charité-Universitätsmedizin, Berlin) and cultured as previously reported.⁶ *CFLAR*^{KO} cell lines were generated using the CRISPR/Cas9 system, either by lentiviral infection or electroporation, with CRISPR RNA CCGGGATGTTGCTATAGAT. Single-cell-derived cell populations were then screened for the deletion of *CFLAR* by using Western blot.

Cell viability assay

A total of 8000 cells were seeded in 96-well plates, and the following day, they were treated as indicated in the figures.

After 24 hours, their viability was assessed using the CellTiter-Glo assay (G7571; Promega, Madison, WI).

Reagents and antibodies

The reagents used included Emricasan (S7775-0005; Absource Diagnostics), GSK'963 (receptor-interacting serine/threonine-protein kinase 1 [RIPK1] inhibitor; S8642; Selleck Chemicals), human TNF, TRAIL, and CD95L (kind gift from H.W.), necrostatin-1 (NSA; S8251; Selleck Chemicals), and LPS (13106-MM; Invivogen).

The antibodies used included caspase-8 (9746; Cell Signaling Technology), tubulin (T9026; Sigma Aldrich), cFLIP (56343; Cell Signaling Technology), RIPK1 (610459; BD Bioscience), Fas-associated death domain protein (immunoprecipitation:14906; Proteintech and immunoblotting: 2782S; Cell Signaling Technology), and TRAIL-R2 (80745; Cell Signaling Technology).

Immunohistochemistry and immunofluorescence

Isolated organs were fixed in 4% paraformaldehyde for 24 hours, washed with phosphate-buffered saline, and embedded in paraffin. Tissues were then serially cut into 2- μ m sections and stained for BCL6 (5650; Cell Signaling), B220 (BD, clone RA3-6B2), CD138 (BD, clone 281-2), and Ki67 (12202; Cell Signaling). GCs were quantified from BCL6 staining using the software QuPath, and Ki-67 positivity was quantified using ImageJ. Cleaved caspase-3 immunofluorescence was performed as previously described.²³

Flow cytometry

Splenocytes from mice aged 30 weeks were stained with 2 different antibody panels comprising the following fluorescently labeled antibodies: BD Biosciences: CD45 (APC-Cy7 and Alexa Fluor700, clone 30-F11), B220 (pacific-blue and BV786, clone RA3-6B2), CD95/Fas (BV510, clone Jo2), immunoglobulin D (IgD) (BUV496, clone 217-170), IgM (BV711, clone II/41), CD19 (BUV395, clone 1D3), CXCR4 (PE, clone 2B11), and CD38 (BUV395, clone 90/CD38); BioLegend: CD3 (PerCP-Cy5.5, clone 281-2), CD138 (PE/Cy7 and PerCP-Cy5.5, clone 281.2) CD21/CD35 (PE/Cy7, clone 7E9), CD23 (BV510, clone B3B4), CD86 (PE/Cy7, clone GL1), and a fixable viability kit (423105; Zombie NIR); and Invitrogen: CD93 (PE, clone AA4.1) and GL-7 (eFluor660, clone GL7). The gating strategies were previously reported.²⁹

Culture and treatment of primary DLBCL cells

Cryopreserved primary DLBCL cells were seeded at 1.5×10^5 cells per well in ultralow attachment plates (Corning, CLS7007) and then centrifuged at 200g for 10 minutes to facilitate spheroid formation. Following overnight incubation, spheroids were treated for 24 hours. Cell viability was assessed by DAPI (4',6-diamidino-2-phenylindole) staining using flow cytometry. The samples included patients 1, 3, and 4 with GCB DLBCL and patient 2 with ABC DLBCL.

Results

cFLIP is overexpressed in DLBCL

First, we assessed expression levels of *CFLAR* in different B-cell lymphomas compared with normal B-cell subsets.³² Transcript abundance of *CFLAR* was significantly increased in all analyzed lymphomas, including DLBCL, compared with naïve and GC B

cells (Figure 1A). We then focused on DLBCL and confirmed elevated cFLIP expression at the protein level in ~60% of tumor specimens derived from 135 patients with DLBCL, when compared with normal B cells from the tonsils (Figure 1B). Of note, patients with DLBCL who showed high *CFLAR* expression exhibited significantly shorter overall survival than patients who showed low *CFLAR* expression⁷ (Figure 1C). When focusing on the ABC and GCB subsets, *CFLAR* mRNA levels were significantly higher in the ABC subtype than the GCB subtype, both in human cell lines (2 different data sets) and in 3 different patient cohort data sets^{5,7,9,35} (Figure 1D-H; supplemental Figure 1A-C, available on the Blood website). Across the 5 different DLBCL clusters, the 2 ABC-enriched genetic subtypes C1 and C5 had a higher abundance of *CFLAR* transcript than the GCB-enriched C3 DLBCL⁵ (Figure 1I). No significant survival differences were observed across all ABC patients (supplemental Figure 1D) or within specific subgroups such as cluster BN2 (analogous to C1) and cluster MCD (analogous to C5; supplemental Figure 1E-F).⁷ Notably, the MCD cluster did not include any *CFLAR*-low patients. This suggested that the survival difference observed in Figure 1C probably reflects the fact that ABC has a poorer prognosis and higher *CFLAR* levels than GCB. Further analysis of specific mutational combinations, including *MYD88* and *CD79B*, *MYD88/CD79B* with *BCL2* upregulation, and *NOTCH2* with *BCL6* upregulation (supplemental Figures 1F, 2A-F), also revealed no significant survival differences based on *CFLAR* expression. However, the only context in which *CFLAR* expression affects survival was in *MYD88*-mutant/*BCL2*-high patients⁷ (supplemental Figure 2G). Altogether, these data indicate that cFLIP is overexpressed in human DLBCL and that the ABC subtype exhibits higher levels of cFLIP, both at the transcriptional and protein levels, than the GCB subtype.

Cflar deletion prevents *Myd88*^{p.L252P} and *BCL2*-driven expansion of the GCs

To study the role of cFLIP in DLBCL in vivo, we used a well-characterized genetically engineered mouse model of DLBCL. The autochthonous *Myd88*^{c.p.L252P/WT} *Rosa26*^{LSL.BCL2.IRES.GFP/WT} *Cd19*^{Cre/WT} (MBC) mouse line is based on a conditional *Myd88*^{p.L252P} allele (orthologous position of human *MYD88*^{p.L265P}), a conditional *BCL2* overexpressing allele (*Rosa26*^{LSL.BCL2.IRES.GFP}), and a *Cd19*^{Cre} recombinase to achieve the expression of the oncogenic mutant of *Myd88* and of the *BCL2* transgene selectively in B cells (Figure 2A).^{29,30,37-40} The MBC mice were previously reported to develop clonal lymphomas that display transcriptomic profiles resembling ABC DLBCL.^{28,30} MBC mice were crossed to mice bearing a conditional *Cflar* KO allele,³¹ *Cflar*^{F/F}, to generate the *Cflar*^{F/F} *Myd88*^{c.p.L252P/WT} *Rosa26*^{LSL.BCL2.IRES.GFP/WT} *Cd19*^{Cre/WT} mice, referred to as FFMBC (Figure 2A). As a control, we used the *Cflar*^{F/F} *Myd88*^{c.p.L252P/c.p.L252P} *Rosa26*^{LSL.BCL2.IRES.GFP/LSL.BCL2.IRES.GFP} mice, referred to as FFMBB. We first assessed the role of cFLIP in B-cell homeostasis. As reported earlier,²⁴ B-cell-specific deletion of *Cflar* (*Cflar*^{F/F} *Cd19*^{Cre/WT} mice, referred to as FFC) did not have pronounced impact on B-cell development and caused only a slight, but significant, reduction of the percentage of total B cells and GC B cells, without affecting the percentage of naïve, mature, follicular, marginal zone B cells and plasma cells (supplemental Figure 3A-D), compared to control mice (*Cflar*^{F/F}, referred to as FF).

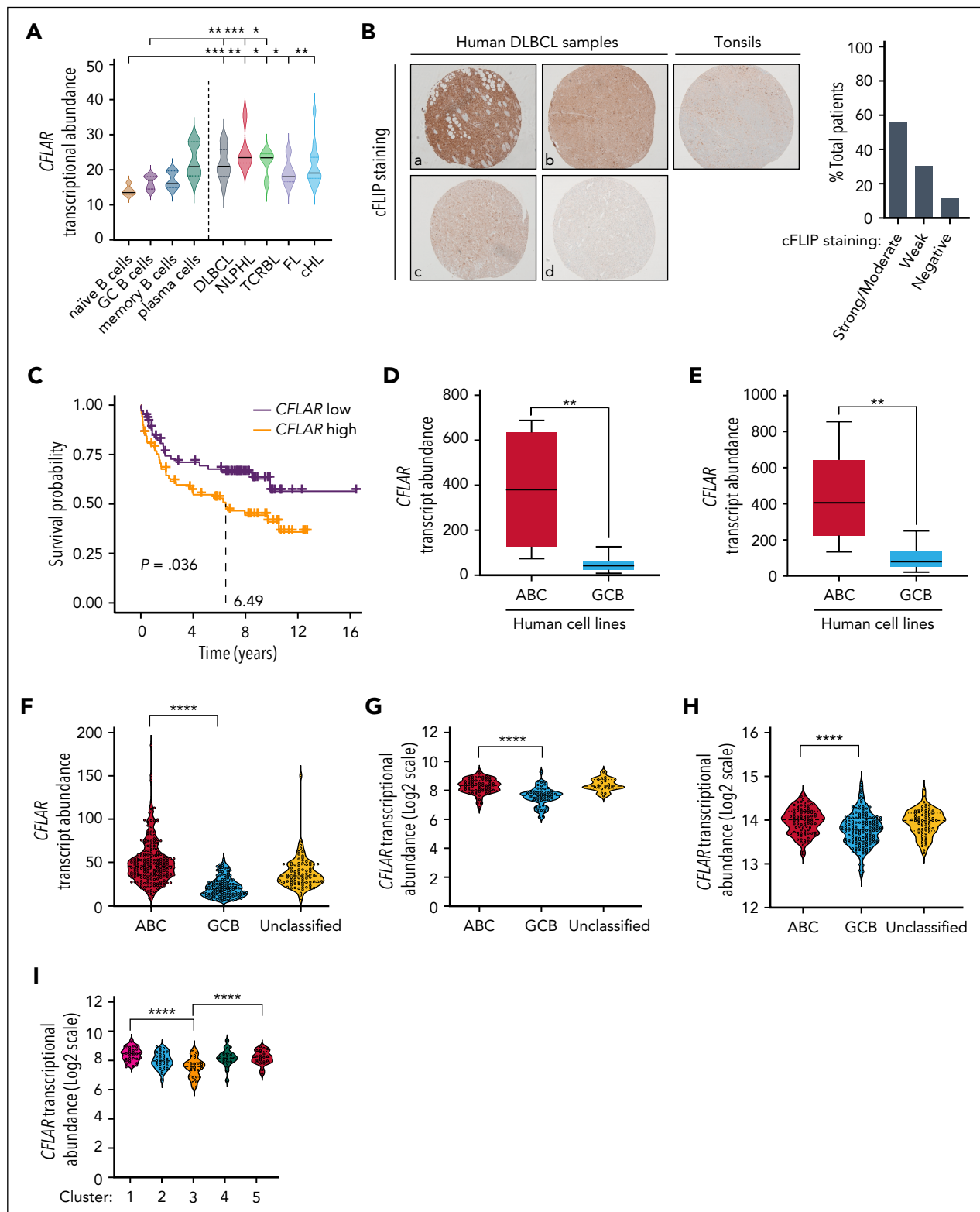


Figure 1. CFLAR is overexpressed in DLBCL. (A) Comparison of CFLAR mRNA expression levels³² between different types of lymphoma and nonmalignant B cells. (B) cFLIP staining in human DLBCL and tonsil (positive control for the staining) tissue sections. The different staining intensity scoring categories are shown (left) (a = strong, b = moderate, c = weak, and d = negative; low magnification). The percentage of the different staining intensity categories over the total number of samples (right). (C) Kaplan-Meier survival curves of patients with DLBCL⁷ who showed high (n = 70; median survival, 6-49 months) or low (n = 70) expression of CFLAR mRNA. The analysis compares the top 30% of the patients with the highest CFLAR expression to the bottom 30% with the lowest expression. (D-E) Box plot showing RNA-seq expression levels of the CFLAR in B-cell lymphoma subtypes categorized into ABC like (red) and GCB like (blue). The 2 different data sets used were from reference³³ (D) and³⁴ (E). Statistical significance was determined using the Mann-Whitney U test. (F-H) CFLAR transcript abundance was assessed in transcriptionally defined ABC, GCB, and unclassifiable DLBCL subtypes, using 3 independent data sets from reference⁷ (F),⁵ (G), and^{9,35} (H). (I) CFLAR RNA expression in the DLBCL genetic subtypes.⁵ * $P \leq .05$; ** $P \leq .01$; *** $P \leq .001$; **** $P \leq .0001$. CHL, classical Hodgkin lymphoma; FL, follicular lymphoma; NLPDL, nodular lymphocytic-predominant Hodgkin lymphoma; TCRBL, T-cell-rich B-cell lymphoma.

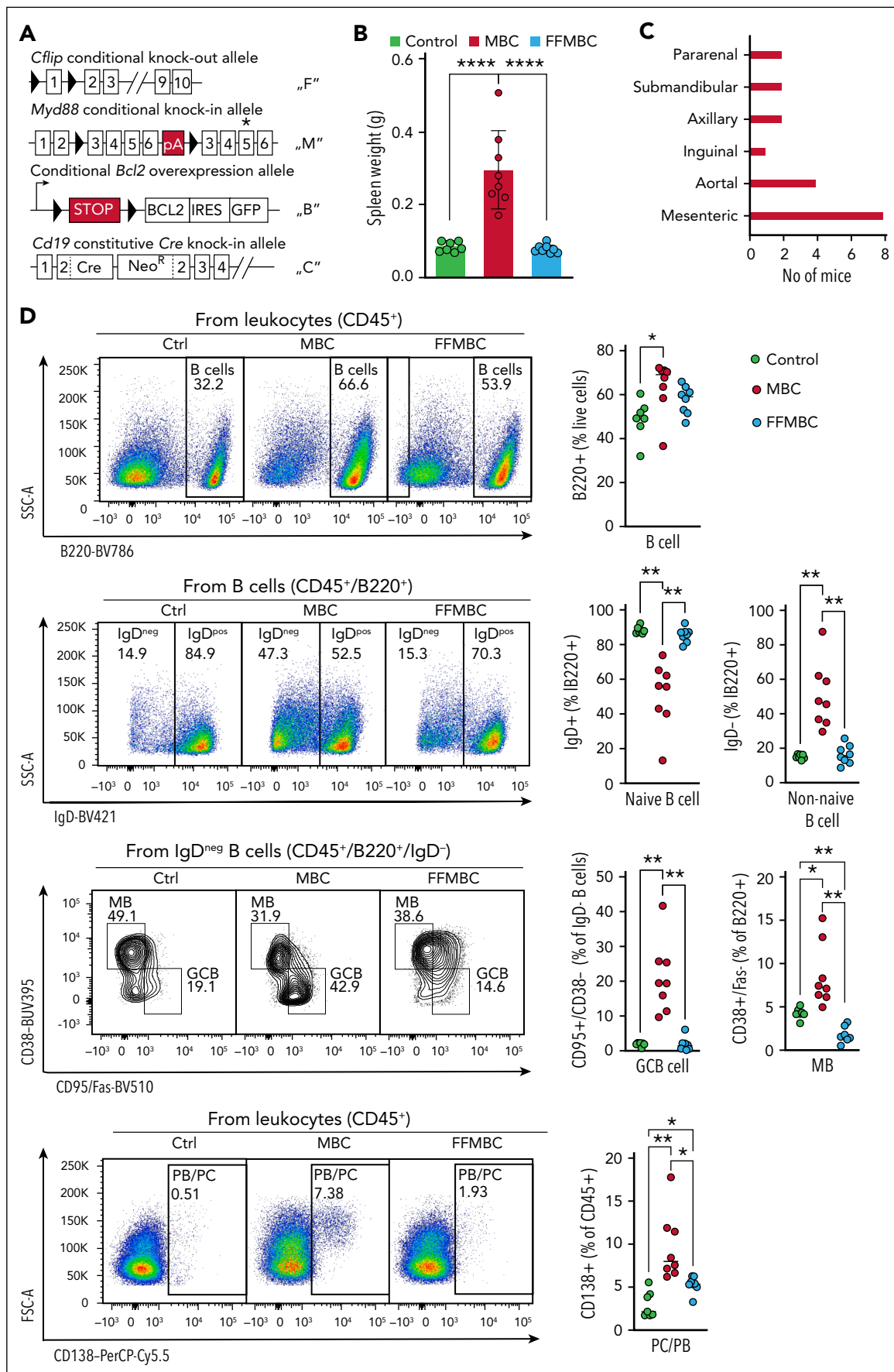


Figure 2. *Cflar* deletion prevents *Myd88*^{L252P} and *BCL2*-driven expansion of CD138⁺ cells. (A) Cartoon depicting the used alleles. Exon 1 of endogenous *Cflar* was flanked by *loxP* sites (triangles). Exons 2 to 6 of endogenous *Myd88* were flanked by *loxP* sites. The asterisk on exon 5 indicates the L252P point mutation. Human *BCL2*

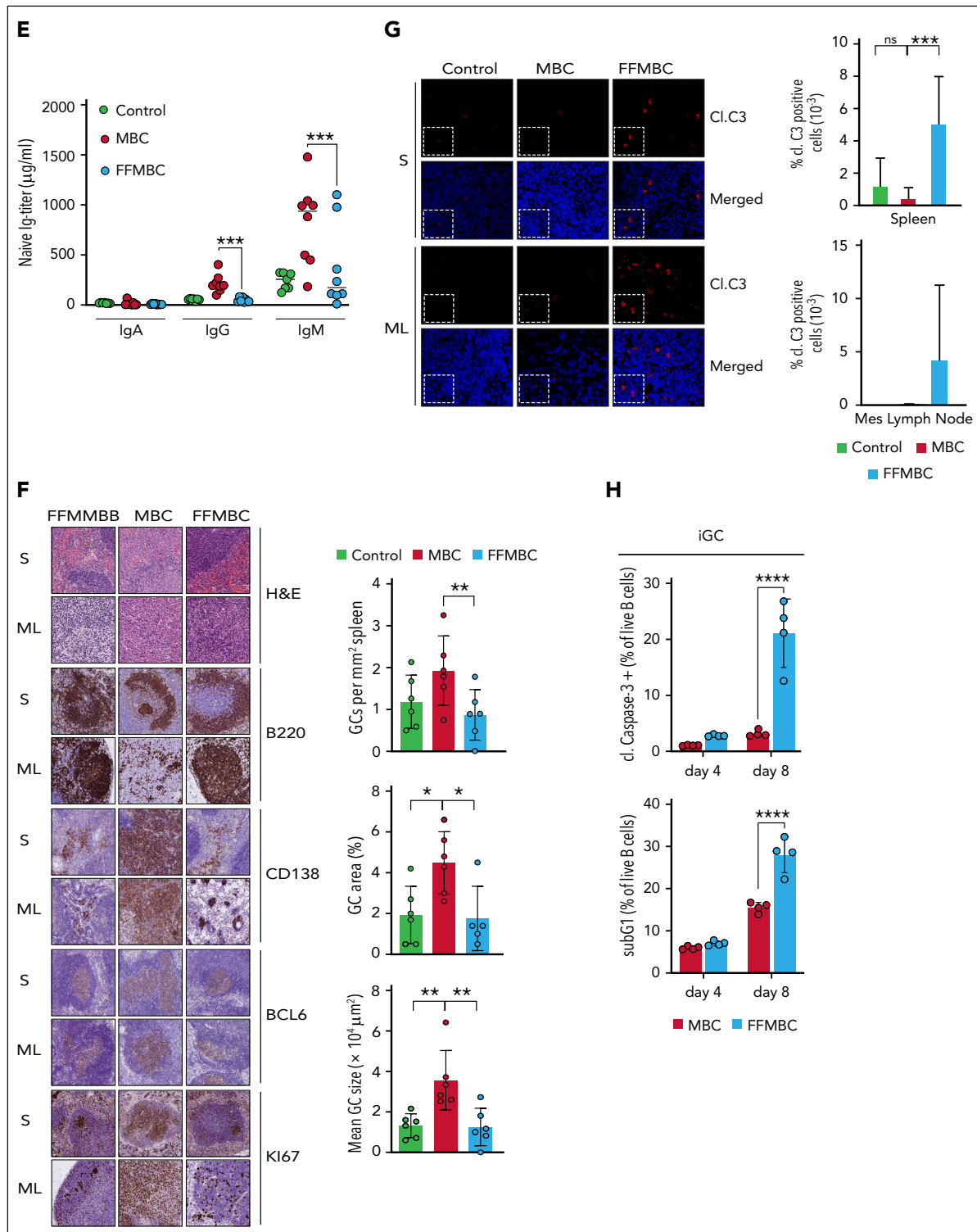


Figure 2 (continued) cDNA expression is prevented by a lox-stop-lox cassette (STOP). EGFP expression is coupled to BCL2 expression by an internal ribosomal entry site. The construct is inserted into the Rosa26 locus. Both alleles have been previously published.²⁸ The *Cd19^{Cre}* allele is a knockin into the *Cd19* locus and has been previously published.²⁸ (B-C) Spleen weight and lymph node enlargement (lymph nodes >10 mm) from mice of the indicated genotypes. Control mice are of the FFMBB genotype. (D) Splenocytes of control, MBC, and FFMBC mice aged 30 weeks (n = 8 per genotype) were analyzed by flow cytometry, and the relative amounts of different subtypes of B cells were quantified. (E) Serum immunoglobulin levels of control, MBC, and FFMBC mice aged 30 weeks (n = 8 per genotype). (F) Spleen and mesenteric lymph node sections from mice aged 30 weeks were stained with the indicated antibodies (top). GC structures stained by BCL6 were quantified with the QuPath software. Control (n = 6), MBC (n = 6), and FFMBC (n = 6; bottom; scale: 0.45 µm/pixel). (G) Spleen and mesenteric lymph node sections from mice aged 30 weeks were stained with a cleaved caspase-3-specific antibody (left). The percentage of cleaved caspase-3-positive cells was quantified in a blind, fully automated fashion using Image J (Cellpose plugin; right; magnification 20x). (H) B cells from MBC and FFMBC mice aged 7 weeks were cocultured for 4 and 8 days with 40LB feeder cells³⁰ and then the percentage of cleaved caspase-3-positive cells and sub-G1 cells was determined by flow cytometry. *P ≤ .05; **P ≤ .01; ***P ≤ .001; ****P ≤ .0001. Cre, cyclization recombination; Cl.C3, cleaved Caspase-3; CTRL, control; EGFP, enhanced green fluorescent protein; FSC-A, forward scatter area; H&E, hematoxylin and eosin; RES, internal ribosomal entry site; LB, CD40 ligand and B-cell activating factor; ML, mesenteric lymph node; neg, negative; PC/PB, plasma cells/plasmablasts; pos, positive; SSC-A, side scatter area.

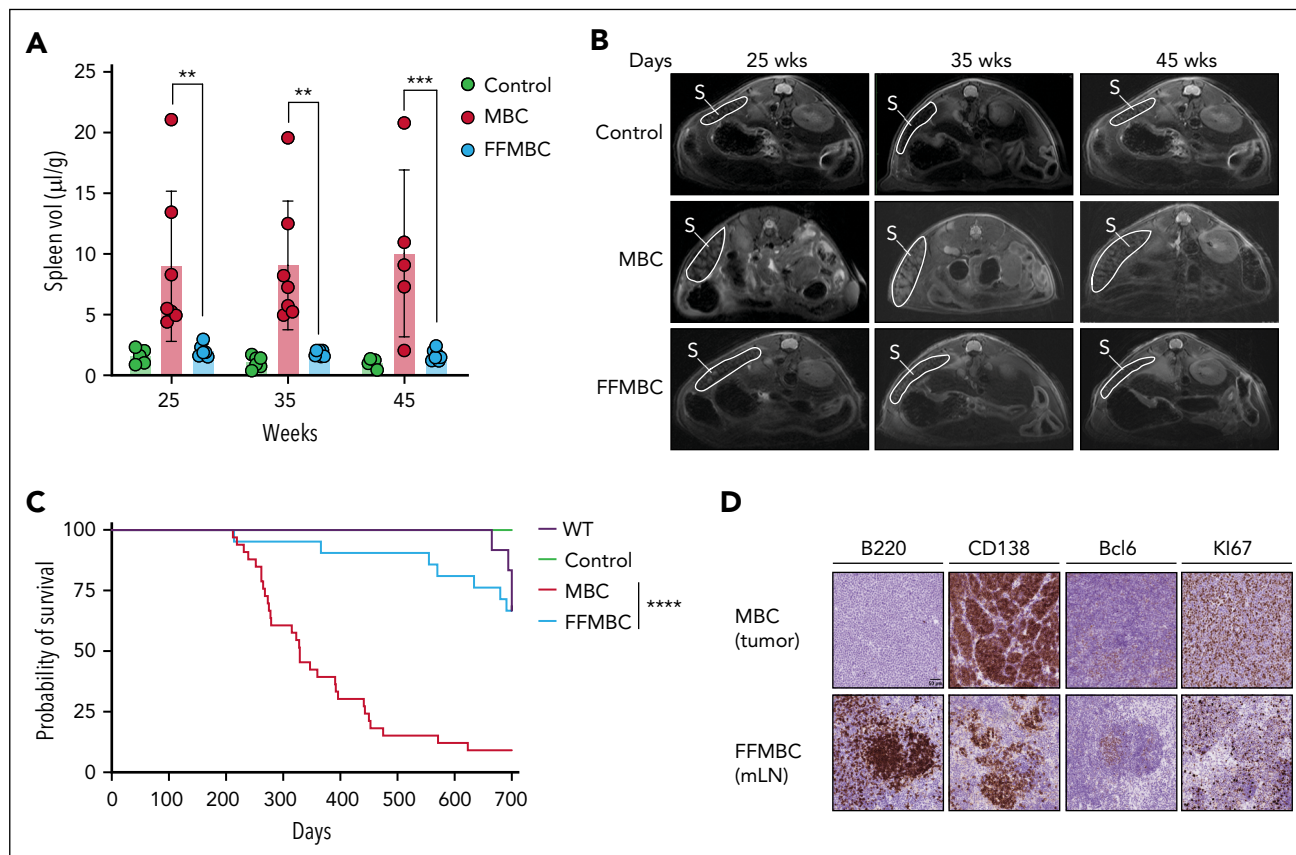


Figure 3. *Cflar* deletion prevents DLBCL development. (A) Spleen volumes of control, MBC, and FFMBC mice of the indicated ages, quantified from magnetic resonance imaging images. Each dot corresponds to 1 mouse. Control mice are of the FFC genotype. (B) Representative magnetic resonance imaging images of FFC, MBC, and FFMBC mice of the indicated ages. (C) Kaplan-Meier survival curves of control (n = 10), MBC (n = 29), FFMBC (n = 21), and WT (n = 12) mice. (D) Representative images of MBC tumor and 20-month-old FFMBC mesenteric lymph node sections, stained with the indicated antibodies. Scale: 0.22 $\mu\text{m}/\text{pixel}$. ** $P \leq .01$; *** $P \leq .001$; **** $P \leq .0001$. mLN, mesenteric lymph node; vol, volume.

As previously reported,³⁰ MBC mice aged 30 weeks exhibited splenomegaly and lymph node hyperplasia, in comparison to control mice (Figure 2B-C; supplemental Figure 3E). This effect was entirely abrogated by concomitant conditional knockout of *Cflar* in FFMBC animals (Figure 2B-C; supplemental Figure 3E). To investigate the composition of splenocytes and spleen architecture, we performed detailed flow cytometric and immunohistochemical analyses of 30-week-old spleens of MBC, FFMBC, and control mice (Figure 2D; supplemental Figure 3C). Immunophenotyping of MBC splenocytes showed, as previously reported,³⁰ an increased percentage of all B cells, with a relative reduction in naïve B cells. As previously described, MBC spleens showed an increase in the GC B-cell population, with a significantly enlarged dark zone and decreased light zone B-cell population, compared to control mice (Figure 2D; supplemental Figure 3F). Furthermore, MBC mice showed a significantly higher number of plasmablasts/plasma cells and a bigger fraction containing memory B cells (Figure 2D). Strikingly, homozygous introduction of the conditional *Cflar* allele into the MBC background led to a complete reversal of the MBC phenotype. Compared to MBC mice, FFMBC spleens harbored a significantly increased number of naïve B cells (Figure 2D). Compared to MBC spleens, the GC B-cell and plasma cell/plasmablast populations were significantly decreased in FFMBC spleens to the level of control mice (Figure 2D). Enzyme-linked immunosorbent assay of sera of FFMBC, MBC, and control mice aged 30

weeks showed significantly increased production of IgG and IgM in MBC. In sharp contrast, FFMBC mice harbored significantly lower immunoglobulin levels, comparable to control mice (Figure 2E). Immunohistochemical stainings of 30-week-old MBC spleens showed an increase in the number and size of GCs and in the number of CD138⁺ cells compared to control and FFMBC mice, based on BCL-6, Ki67, and CD138 stainings, respectively, further supporting the flow cytometric data (Figure 2F).

To determine whether cell death plays any role in the aforementioned *Cflar* deletion-mediated phenotype, we stained spleen and lymph node sections with a cleaved caspase-3-specific antibody. We detected significantly higher levels of cleaved caspase-3 in FFMBC mice with respect to control and MBC mice (Figure 2G). To confirm that cell death occurred during the GC reaction, we took an ex vivo approach called induced GC³⁶ culture, using B cells isolated from MBC and FFMBC mice (supplemental Figure 3G). We observed that the genetic deletion of *Cflar* caused a reduction of the total B-cell fraction in culture and B-cell activation, together with a reduction of IgG1-expressing B cells and diminished IgG1 secretion (supplemental Figure 3H). A significantly higher fraction of caspase-3-positive cells and a bigger sub-G1 population (Figure 2H) indicated that the deletion of *Cflar* renders *Myd88* mutant, *BCL2* overexpressing GC B cells more prone to apoptotic cell death. Taken together, these data suggest that

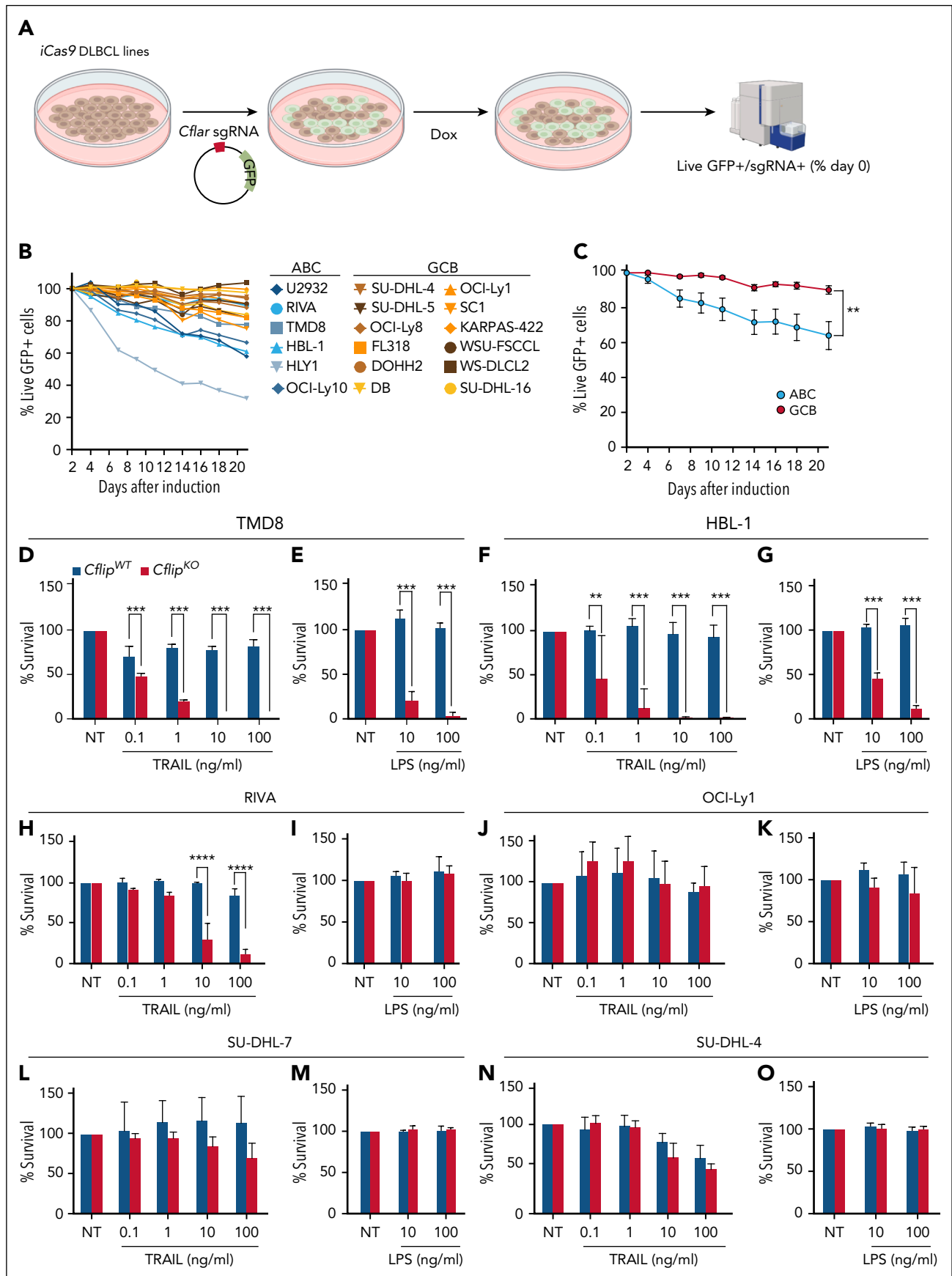


Figure 4. CFLAR deletion sensitizes ABC-DLBCL but not GCB-DLBCL cells to TRAIL- and LPS-induced apoptosis. (A) Cartoon depicting the CRISPR-mediated cell competition assay. (B) The survival of the CFLAR gRNA/GFP-positive cells of the indicated cell lines was monitored over time by flow cytometry. (C) Mean survival values for

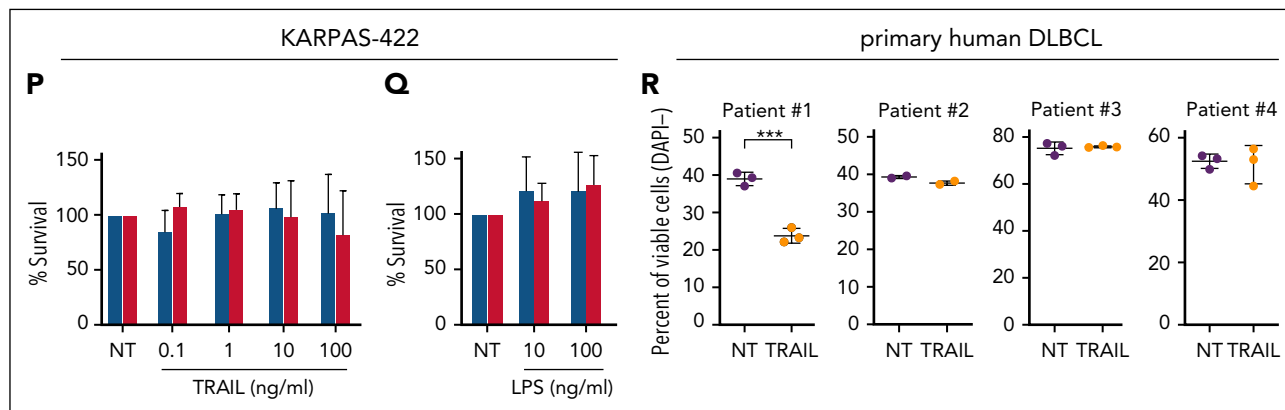


Figure 4 (continued) the ABC- and GCB-DLBCL lines shown in panel B, $\pm$ standard error of the mean. (D-Q) *CFLAR* WT and KO ABC-DLBCL and GCB cells were treated with TRAIL and LPS at the indicated concentrations. A cell viability assay was carried out after 24 hours of treatment. Data are presented as means $\pm$ standard deviation ($n = 3$). (R) Primary human DLBCL cells from 4 different patients were cultured for 24 hours and then treated with 100 ng/mL TRAIL for a further 24 hours. The percentage of viable cells was then measured by flow cytometry. $**P \leq .01$, $***P \leq .001$. Dox, doxycycline; NT, untreated; sgRNA, single-guide RNA.

cFLIP expression in B cells is essential to prevent cell death during the GC reaction and allows the expansion of post-GC B cells induced by oncogenic *Myd88* and *BCL2* overexpression.

***Cflar* deletion prevents DLBCL development**

Next, we sought to determine the effect of the B-cell-specific deletion of *Cflar* on DLBCL development and overall survival of MBC mice. For this purpose, we subjected FFC, MBC, and FFMBC mice to longitudinal magnetic resonance imaging scans beginning at the age of 15 weeks. MBC mice were previously shown to develop splenomegaly throughout their lives.³⁰ As shown in Figure 3A-B, *Cflar* deletion completely prevented splenomegaly, with FFMBC mice showing spleen volumes largely similar to control mice (FFC). MBC mice have been previously reported to develop lymphoma with a median survival of 42.3 weeks. Consistent with this earlier finding and with the fact that MBC mice display splenomegaly and plasmablast accumulation, we observed that MBC mice developed lymphoma and had a median survival of 47 weeks (Figure 3C). These MBC lymphomas showed the reported expression pattern of CD138 and a high Ki67 index (Figure 3D). In sharp contrast, we did not detect any signs of lymphomagenesis in FFMBC mice (Figure 3C-D). Of note, FFMBC mice also showed survival similar to wild-type (WT) mice (Figure 3A).

Taken together, *Cflar* deletion prevented lymphoma development in the MBC mouse model in the analysis time window of 100 weeks. Therefore, our data support the notion that cFLIP expression in B cells is necessary to sustain oncogenic *Myd88*- and *BCL2*-driven DLBCL development in our model system.

***CFLAR* deletion sensitizes ABC-, but not GCB-DLBCL cells, to TRAIL- and LPS-induced apoptosis**

To determine the dependence of human DLBCL cells on cFLIP, we used a CRISPR-mediated cell competition assay using a panel of 18 lines, including 6 of the ABC and 12 of the GCB subtype, with different mutational backgrounds, among which we monitored the survival of the *CFLAR* guide RNA-transduced cells over time (Figure 4A; supplemental Figure 4A-B). Importantly, only ABC-DLBCL lines showed dependence on cFLIP,

independent of the *MYD88* and *CD79B* mutational background and *BCL2* expression levels (Figure 4B-C).

cFLIP prevents apoptosis downstream of different immune receptors that play an important role in B-cell biology, including DRs TNFR1, CD95 and TRAIL-Rs, and the toll-like receptor 4 (TLR4). In order to verify whether cFLIP is necessary to ensure survival of DLBCL cells following the engagement of any of the aforementioned receptors, we stimulated 3 ABC-DLBCL lines (TMD8 and HBL-1, with high *CFLAR* expression levels, and RIVA, with low *CFLAR* expression levels; supplemental Figure 1A-B) and 4 GCB-DLBCL lines (SU-DHL-4, SU-DHL-7, OCI-Ly1, and KARPAS-422) with their cognate ligands. CRISPR/Cas9-mediated deletion of *CFLAR* (supplemental Figure 4C-I) significantly impaired the viability of TMD8, HBL-1 and RIVA after TRAIL stimulation and of TMD8 and HBL-1 following LPS stimulation in a dose-dependent manner, whereas the viability of cFLIP proficient cells was not affected by TRAIL treatment or LPS (Figure 4D-I). Importantly, deletion of *CFLAR* in the GCB-DLBCL lines did not cause loss of viability upon TRAIL or LPS treatment (Figure 4J-Q). The absence of cFLIP did not impact the survival of ABC- and GCB-DLBCL cells treated with TNF and CD95L (supplemental Figure 4J-W), with the exception of SU-DHL-7, which exhibited reduced viability after CD95L treatment (supplemental Figure 4U). Notably, the fact that TMD8, HBL-1 and RIVA bear *BCL2* gene amplification indicates that *BCL2* overexpression cannot prevent TRAIL-mediated death when cFLIP is absent. This, in turn, indicates that ABC-DLBCL cells are type I cells because they can undergo TRAIL-induced apoptosis independent of the mitochondria and therefore independent of the intrinsic pathway. Furthermore, *CFLAR* deletion did not enhance the killing activity of the BH3 mimetic ABT-737 and the Bruton tyrosine kinase inhibitor, ibrutinib (supplemental Figure 4X-Y). Finally, we treated human primary DLBCL cells from 4 different patients with TRAIL (supplemental Figure 4Z) and observed that 1 patient showed a modest but significant response to TRAIL-induced loss of survival (Figure 4R). Taken together, these results show that ABC-DLBCL cells have a significantly higher dependence on cFLIP than the GCB

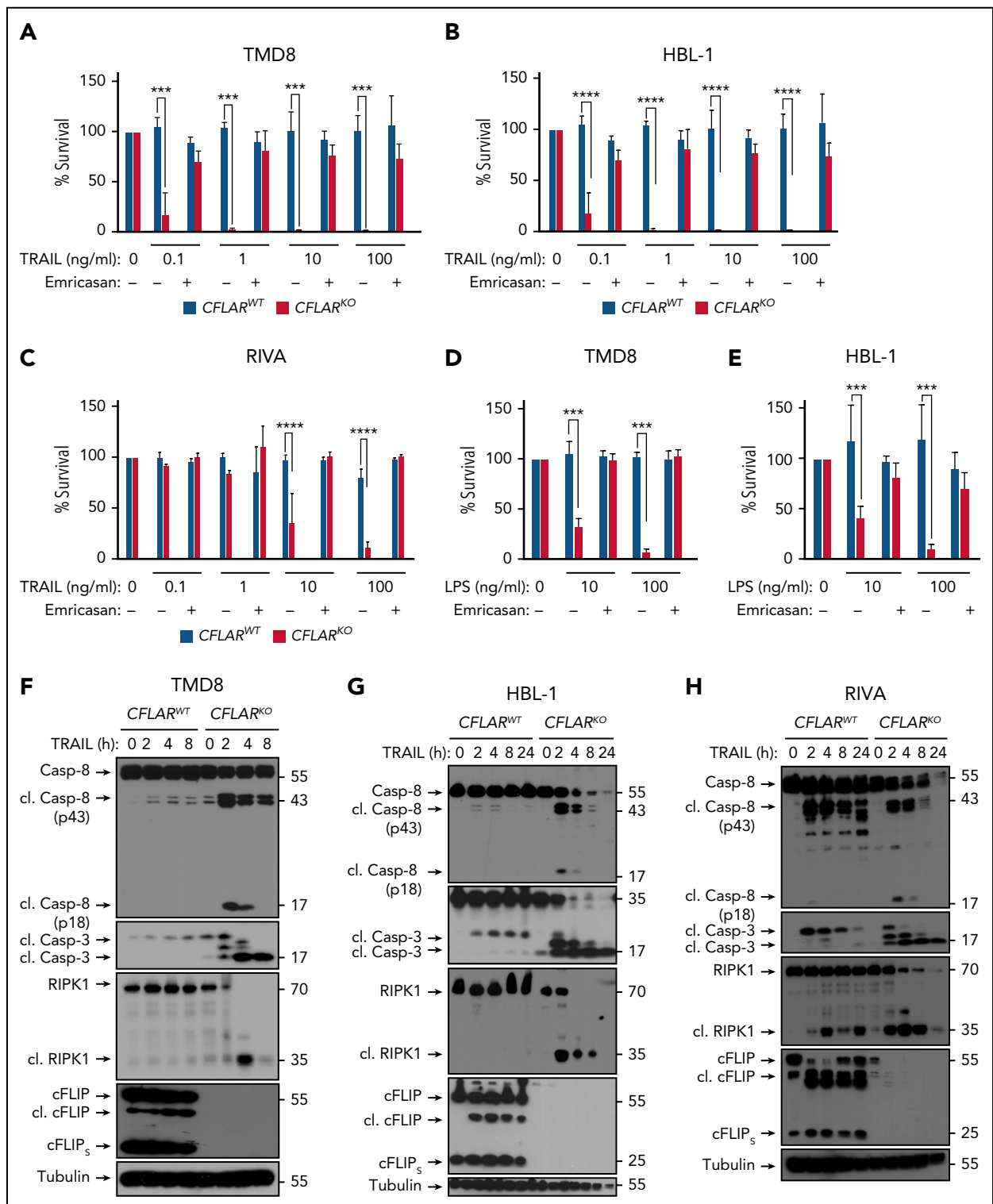


Figure 5. CFLAR deletion enhances caspase-8 activation in ABC-DLBCL cells. (A-E) CFLAR WT and KO TMD8, HBL-1, and RIVA were treated with the indicated concentration of TRAIL or LPS in the presence or absence of the pan-caspase inhibitor emricasan (E; 1 μ M). A cell viability assay was carried out after 24 hours. Data are presented as means $\pm$ standard deviation (n = 3). (F-J) CFLAR WT and KO TMD8, HBL-1, and RIVA were treated with TRAIL (10 ng/mL) or LPS (10 ng/mL) for the indicated time points, and protein extracts were immunoblotted with the indicated antibodies. (K) CFLAR WT and KO TMD8 were treated with FLAG-tagged TRAIL (500 ng/mL) for 20 minutes, and the TRAIL complex was immunoprecipitated by anti-FLAG beads. Immunocomplexes were analyzed by immunoblotting using the indicated antibodies. (L) CFLAR WT and KO HBL-1 were treated with FLAG-TRAIL (500 ng/mL) for the indicated time point, and the TRAIL complex was immunoprecipitated by anti-FLAG beads. TRAIL complex-depleted lysates were then subjected to immunoprecipitation by an anti-FADD antibody. Immunocomplexes were analyzed by immunoblotting using the indicated antibodies. (M) Primary human DLBCL cells from 4 different patients were cultured for 24 hours and then treated with TRAIL (100 ng/mL) for 4 hours. Protein extracts were immunoblotted with the indicated antibodies. FADD, Fas-associated death domain protein.

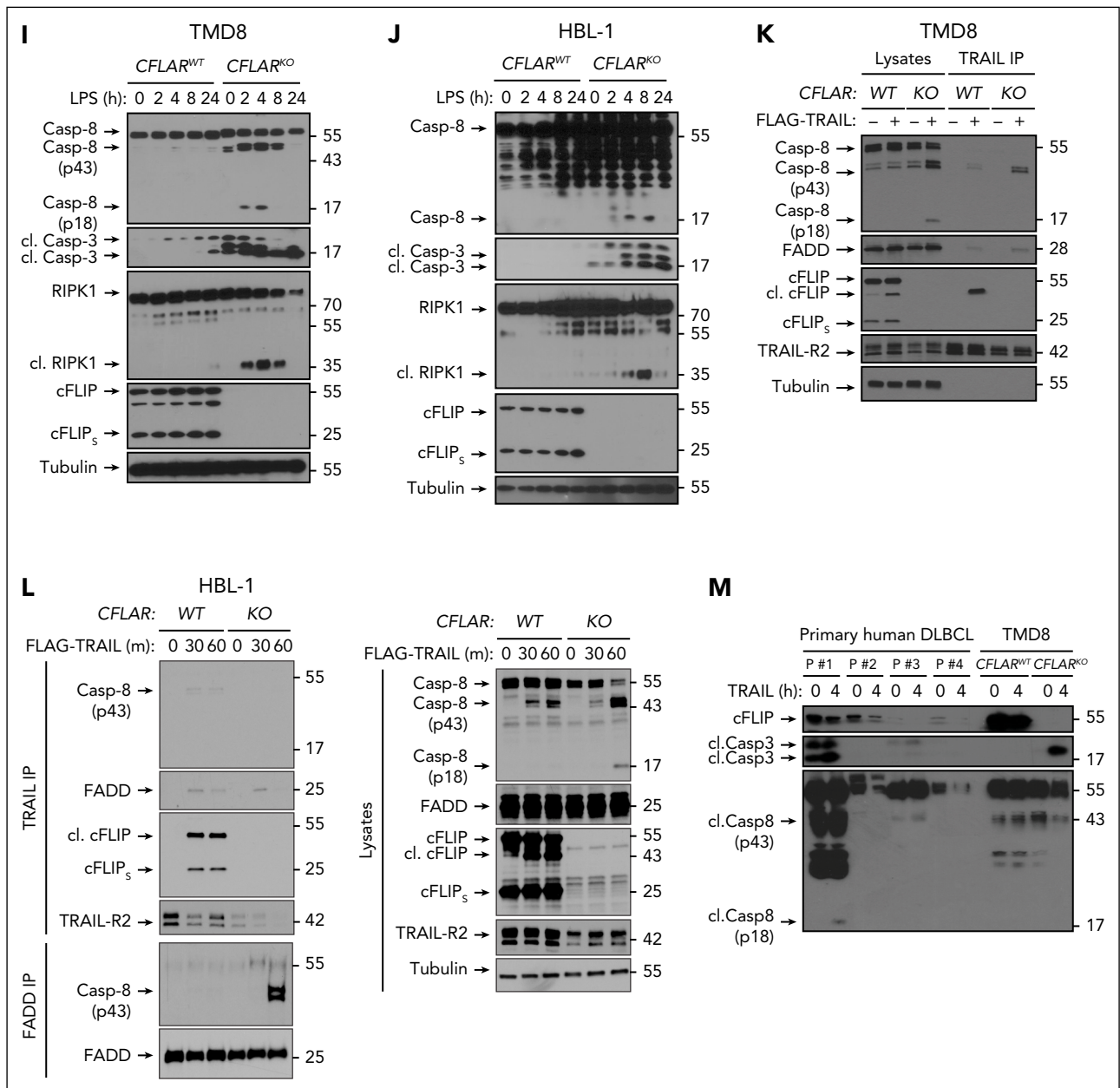


Figure 5 (continued)

subtype. Moreover, *CFLAR* deletion promotes TRAIL- and LPS-mediated death in ABC-DLBCL cells but not GCB-DLBCL cells.

***CFLAR* deletion enhances caspase-8 activation in ABC-DLBCL cells**

Deletion of *Cflar* has been reported to cause deregulation of both apoptosis and necroptosis.¹⁹ To investigate the mechanisms by which the absence of cFLIP impairs the viability of ABC-DLBCL cells, we treated *CFLAR*^{WT} and *CFLAR*^{KO} TMD8, HBL-1 and RIVA cells with TRAIL or LPS in the presence of the pan-caspase inhibitor emricasan, the RIPK1-specific inhibitor GSK'963 and the necroptosis inhibitor NSA. Although emricasan fully prevented TRAIL- and LPS-mediated loss of viability in *CFLAR*^{KO} cells (Figure 5A-E), GSK'963 and NSA did not provide any protection (supplemental Figure 5A-J). This

suggested that the loss of viability observed in cFLIP-deficient cells has to be ascribed to RIPK1-independent apoptosis. The occurrence of apoptosis was further confirmed by the presence of cleaved caspase-8 and caspase-3 in lysates from TRAIL- and LPS-treated *CFLAR*^{KO} cells (Figure 5F-J), whereas no cleavage was observed in cFLIP-deficient GCB cells (supplemental Figure 5K). In addition, we detected caspase-8 activation at the level of TRAIL-induced complex I and complex II (Figure 5K-L) in *CFLAR*-deficient TMD8 and HBL-1 cells, respectively, but not in *CFLAR* WT cells. Lysates from TRAIL-treated primary human DLBCL cells showed cleavage of caspase-8 and caspase-3 only in 1 patient's sample, patient 1 (Figure 5M), consistent with the fact that cells from this patient exhibited a moderate reduction of viability after TRAIL treatment. Intriguingly, cells from patient 1 also exhibited the highest levels of cFLIP, suggesting that the therapeutic

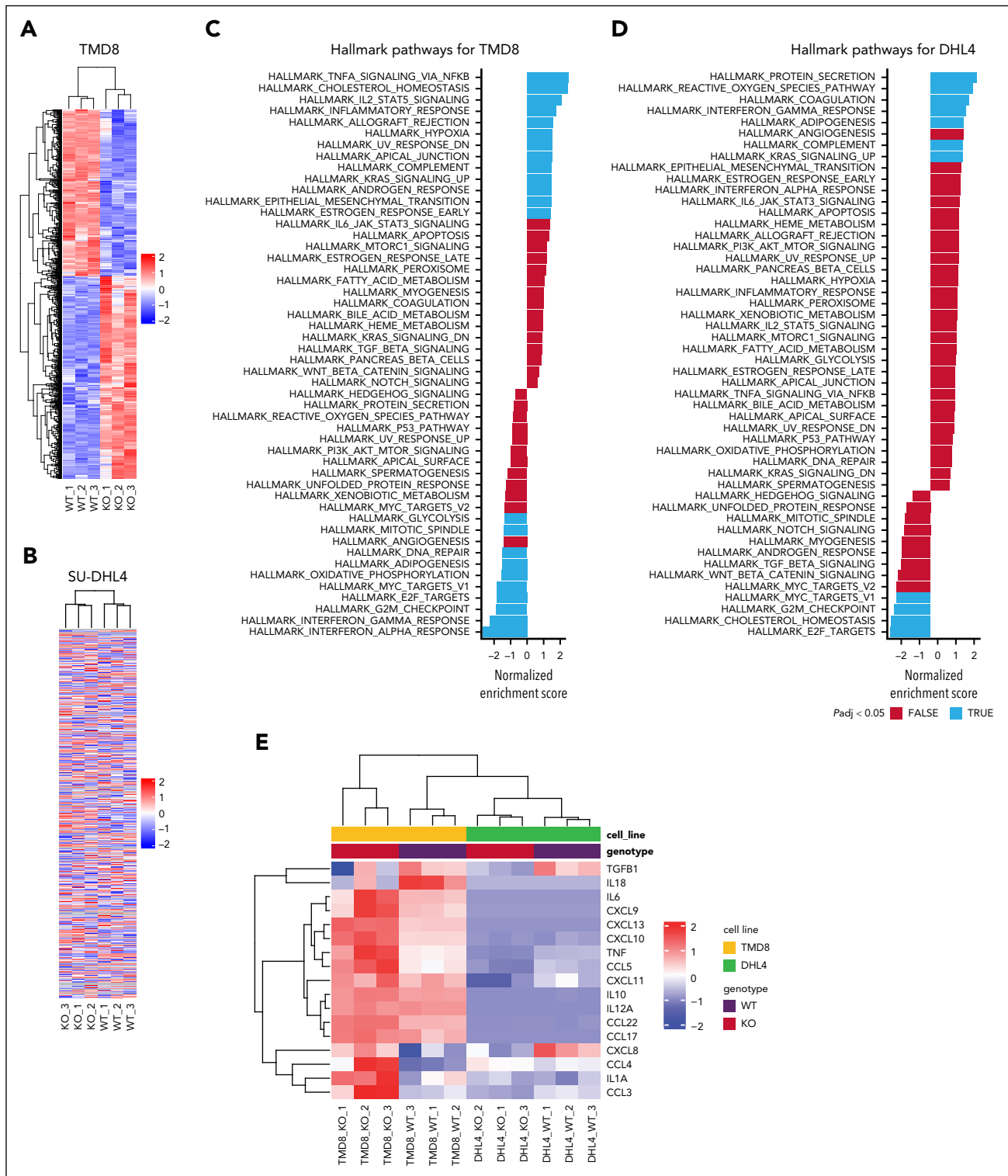


Figure 6. cFLIP suppresses inflammatory gene expression. (A) Heat maps of the top 1030 DEGs determined by RNA-seq analysis in *CFLAR*-deleted TMD8 compared with their *CFLAR* WT counterparts. (B) Heat map showing the expression levels of the DEGs found in panel A in SU-DHL-4 *CFLAR* KO compared with the *CFLAR* WT. Panel B contains the same row order of hierarchical clustering shown in panel A. (C-D) Gene set enrichment analysis of hallmark gene sets in TMD8 and SU-DHL-4 cells. A positive normalized enrichment score indicates gene sets enriched in the KO genotype. True indicates pathways enriched with $P_{adj} < 0.05$. (E) Heat map of the mRNA expression of selected cytokine-related genes involved in inflammatory response, in *CFLAR* WT and KO TMD8 and SU-DHL-4. DEG, differentially expressed genes; IL2, interleukin 2; P_{adj} , BH-adjusted P value; TGF, transforming growth factor.

targeting of cFLIP in high cFLIP-expressing DLBCL tumor cells might further sensitize them to TRAIL stimulation and promote their elimination.

Altogether, we conclude that cFLIP prevents apoptosis in ABC-DLBCL cells but not GCB-DLBCL cells by preventing caspase-8 activation after TRAIL-R1/2 and TLR4 stimulation.

cFLIP suppresses inflammatory gene expression

The cFLIP/caspase-8 heterodimer is not only involved in cell death regulation but also in cytokine and chemokine production downstream of NF- κ B.^{41–43} Because ABC-DLBCL cells are characterized by chronic active B-cell receptor signaling that results in constitutive NF- κ B activation, we next wanted to determine whether *CFLAR* deletion had an impact on the transcriptional activation of proinflammatory factors. To this purpose, we performed RNA-seq of *CFLAR*^{WT} and *CFLAR*^{KO} TMD8 (ABC-DLBCL). As a control, we used the GCB-DLBCL line SU-DHL-4. *CFLAR* deletion in TMD8 caused a pronounced gene expression change (Figure 6A) that was substantially different from the change observed in SU-DHL-4 (Figure 6A-B; supplemental Figure 5A-C). Gene set enrichment analysis showed that inflammatory gene sets, including NF- κ B target genes, were among the most expressed genes in *CFLAR*-deficient TMD8 cells but not in SU-DHL-4 cells (Figure 6C-D). Consistent with this, the expression of known NF- κ B target cytokines and chemokines was increased in *CFLAR*^{KO} TMD8 compared with *CFLAR*^{WT} TMD8. However, deletion of *CFLAR* in SU-DHL-4 did not lead to the transcriptional activation of these NF- κ B target genes (Figure 6E). Based on this evidence, we concluded that cFLIP suppresses proinflammatory gene expression in ABC-DLBCL cells but not GCB-DLBCL cells in a cell death-independent manner.

Discussion

In this study, we demonstrated in an autochthonous murine model that cFLIP expression in B cells is essential for *Myd88*/*BCL2* overexpression-driven lymphomagenesis. *BCL2* overexpression, which causes inhibition of intrinsic apoptosis, is detected in human follicular lymphoma, DLBCL, chronic lymphocytic leukemia, and mantle cell lymphoma.^{44,45} In mice, *BCL2* overexpression significantly accelerates *c-Myc*-driven lymphoma and oncogenic *Myd88*-driven DLBCL.^{28,46} Despite the clear involvement of intrinsic apoptosis inhibition in lymphoma development, much less is known about the role of extrinsic apoptosis in lymphomagenesis. In particular, it is unclear whether suppression of extrinsic apoptosis would be necessary to support lymphomagenesis in settings where the intrinsic pathway is already suppressed or whether intrinsic apoptosis suppression would be sufficient to promote lymphoma and, therefore, bypasses the need for concomitant inhibition of extrinsic apoptosis. By deleting *Cflar* specifically in B cells, we showed that the extrinsic apoptosis pathway counteracts DLBCL pathogenesis, despite the inactivation of intrinsic apoptosis. Based on the published role of cFLIP during the GC reaction, on the presence of cleaved caspase-3-positive cells in spleen and lymph nodes and ex vivo expanded B cells, and the relative reduction of class-switched cells in FFMBC compared with MBC splenocytes, we assume that the lack of lymphoma development upon loss of *Cflar* is because of

a cell death-dependent, most probably apoptotic, block in the GC reaction.

Previous reports indicate that GC B cells express the highest levels of TRAIL-Rs and CD95, 2 of the most characterized DRs.⁴⁷ Given the fact that we detected specific cleaved caspase-3 staining in spleen sections of FFMBC mice aged 30 weeks, and that these mice did not develop lymphoma, we speculated that cFLIP is important to guarantee (pre)malignant cell survival during the GC reaction, that in the case of the MBC mice is induced by oncogenic mutant *Myd88*, and prevent cell death downstream of either CD95 or TRAIL-Rs, or both. Elevated levels of cFLIP have been reported in different types of NHLs, including DLBCL, and seem to correlate with reduced patient survival.^{26,27} The ABC subtype exhibits significantly higher levels of cFLIP, both at the mRNA and protein levels, than the GCB subtype, both in patients and cell lines. Consistently, ABC-DLBCL cells display significantly higher survival dependence on cFLIP than GCB-DLBCL cells. Furthermore, *CFLAR* deletion sensitizes ABC-DLBCL cells but not GCB-DLBCL cells to TRAIL- and LPS-mediated apoptosis, but not to CD95L- and TNF-mediated apoptosis. These findings suggest that ABC-DLBCL cells are addicted to cFLIP for their survival, and cFLIP protects them from TRAIL pathway stimulation. Interestingly, high cFLIP-expressing primary human DLBCL cells were partially sensitive to TRAIL-induced loss of viability and caspase activation. In line with the findings from cell lines, this strongly hints at the fact that targeting cFLIP in high cFLIP-expressing DLBCL cells would have a therapeutic effect by lowering the threshold of cell death sensitivity.

Most ABC-DLBCL cells, including those we used in this study, have *BCL2* copy number gains and therefore overexpress *BCL2* protein. Our results show that reactivation of the extrinsic apoptotic pathway through *CFLAR* deletion efficiently kills ABC-DLBCL cells despite the overexpression of *BCL2* and the resulting inhibition of intrinsic apoptosis. Therefore, the extrinsic apoptotic pathway can be executed independently of the mitochondria, suggesting that ABC-DLBCL cells are type I cells.²⁰

Apart from protecting cells from cell death, cFLIP also suppresses the expression of NF- κ B-driven inflammatory genes in ABC-DLBCL cells, supporting a cell death-independent role of cFLIP in these tumor cells. We speculate that, by preventing proinflammatory gene expression, cFLIP may contribute to maintaining DLBCL in a cold state, allowing malignant cells to escape immune system surveillance. Therefore, in addition to sensitizing cells to apoptosis, the absence of cFLIP could also enhance the immunogenicity of DLBCL cells.

Together, these findings identify cFLIP as a novel vulnerability in ABC-DLBCL. Furthermore, they provide the rationale for the development of specific cFLIP inhibitors for the treatment of ABC-DLBCL, either as a single agent or in combination with sublethal concentrations of TRAIL-R agonists.

Acknowledgments

The authors thank the Center for Molecular Medicine Cologne Microscopy in-house service for providing access to the Leica STELLARIS 5 confocal microscope. The acquisition of this instrument was supported by the Deutsche Forschungsgemeinschaft (DFG, German Research

Foundation; DFG-INST 216/1174-1 Förderprogramm Forschungs-großgeräte). They also thank Alexandra Florin from the Institute of Pathology for their invaluable technical support. Flow cytometry analyses were supported by the Fluorescence-Activated Cell Sorting Core Facility of the Max Planck Institute for Biology and Ageing, Cologne, Germany. The authors thank the Cologne Center for Genomics for performing the RNA sequencing.

This work was funded by research funding from Center for Molecular Medicine Cologne junior research group program, DFG grants AN 1717/1-1 and 1717/1-2, DFG grant 455784452 as part of the CRC1530 (A05), the Jürgen Manchot Foundation, and The Ministry of Culture and Science of North Rhine-Westphalia (NW21-062A CANTAR) (A.A.); DFG grant 413326622 as part of CRC1399 (A01), grant 424228829 as part of CRC1430 (A09), grant 455784452 as part of CRC1530 (A01), grant RE 2246/17-1, the German Ministry of Education and Research (Bundesministerium für Bildung und Forschung e:Med Consortium InCa) grants 01ZX1901 and 01ZX2201A, the Ministry of Culture and Science of North Rhine-Westphalia (NW21-062A CANTAR), The German Cancer Aid grants 1117240 and 70113041 and the Cancer Aid excellence Program (H.C.R.); José Carreras Leukemia Foundation (DJCLS 02FN/2021) (J.L.), Max Planck Society, DFG grants 496650118, 5504 FOR JA 2439/5-1 TP05, JA2439/4-1, grant 455784452 CRC1530 (C02), and the Ministry of Culture and Science of North Rhine-Westphalia (NW21-062A CANTAR) (R.D.J.); and the Else Kröner-Fresenius-Stiftung (2024_EKEA.21) (R.F.).

Authorship

Contribution: A.A. supervised the study and wrote the manuscript; A.A. and K.T.B. conceptualized and designed the experiments, performed and analyzed most of the experiments, and prepared the figures; D.P.S., K.N., and G.G. helped with the generation of *CFLAR*^{KO} diffuse large B-cell lymphoma (DLBCL) cells; S.S. performed TRAIL-induced complex I and complex II isolation; H.W. provided the tools and expertise to isolate TRAIL-induced complex I and complex II and also provided human recombinant TRAIL and CD95L; J.L., N.S., and B.C. provided reagents, analyzed *CFLAR* transcript abundance in human lymphomas and lymphoma cell lines and the somatic copy number alterations of BCL2 in human lymphoma lines, and assisted with the interpretation and writing of the data respective to the human data; S. Spöck, V.L., and A.V. performed the induced GC reaction experiment and analyzed the data; J. V. and M.P. performed the analysis of the RNA-sequencing (RNA-seq) data and built the relative graphics; R.B. provided the resources for the preparation of mouse tissue sections and for immunohistochemistry; H.C.R. provided the MBC mouse line. S.K. and G.K. performed bioinformatic analysis on the human DLBCL RNA-seq data; A.L. and R.F. designed, performed and analyzed all the fluorescence-activated cell sorting (FACS) experiments and built the relative graphics; R.D.J. designed and analyzed FACS experiment data; M.R. and R.D.J. provided the primary human DLBCL samples; G.K. provided guidance with the MBC model; R.F. and G.K. provided relevant intellectual input; R.E.P. and S. Scheich designed and carried out the CRISPR-mediated cell competition assay and analyzed the data;

T.O. and J.C.E. analyzed *CFLAR* transcript abundance in activated B-cell-like- and germinal center B-cell-like-DLBCL subtypes; L.D.S., P.-H.N., and M.H. provided expertise with primary human DLBCL cultures and analyzed their viability following TRAIL treatment; R.F. assisted with the manuscript writing and data interpretation relative to the MBC model; and all authors read and edited the manuscript and agreed to the final form.

Conflict-of-interest disclosure: H.C.R. received consulting and lecture fees from Roche, Novartis, Takeda, Janssen, BeiGene, Lilly, AbbVie, AstraZeneca, Vertex, and Merck; received research funding from AstraZeneca and Gilead Pharmaceuticals; and is a cofounder of CDL Therapeutics GmbH. R.B. received honoraria for lectures and advisory boards from AbbVie, Amgen, AstraZeneca, Bayer, Bristol Myers Squibb, Boehringer-Ingelheim, Illumina, Janssen, Lilly, Merck-Serono, Merck Sharp & Dohme, Novartis, Qiagen, Pfizer, Roche, Sanofi, Targos MP Inc and is a co-owner of Gnothis Inc (Sweden) and Timer Therapeutics GmbH&Co KG (Deutschland). The remaining authors declare no competing financial interests.

ORCID profiles: K.T.B., 0000-0002-3785-6887; S.S.-S., 0000-0003-0325-1238; D.P.S., 0000-0002-9217-5582; S. Spöck, 0009-0005-0956-0316; L.D.S., 0009-0005-8510-6035; N.S., 0009-0004-3668-507X; J.V., 0000-0002-5330-7167; K.N., 0000-0002-3063-4827; G.G., 0009-0008-3375-4577; A.L., 0009-0000-2891-7038; M.R., 0000-0002-4160-7664; S.K., 0000-0002-2188-9377; H.K., 0000-0003-2796-1429; M.P., 0000-0002-5243-5503; B.C., 0000-0002-6485-8773; G.K., 0000-0001-8395-3701; V.L., 0000-0001-7538-1520; P.-H.N., 0000-0002-3249-7264; R.D.J., 0000-0001-9522-7061; M.H., 0000-0002-7425-4455; H.C.R., 0000-0001-5706-9349; A.A., 0000-0002-0346-4340.

Correspondence: Alessandro Annibaldi, Center for Molecular Medicine Cologne, University of Cologne, Robert-Koch-Str. 21, Cologne 50931, Germany; email: a.annibaldi@uni-koeln.de.

Footnotes

Submitted 21 January 2026; accepted 16 March 2026; prepublished online on *Blood* First Edition 23 April 2026. <https://doi.org/10.1182/blood.2026033320>.

Original data and reagents are available from the corresponding author, Alessandro Annibaldi a.annibaldi@uni-koeln.de, on request. All RNA-sequencing data generated in this study were deposited in the Gene Expression Omnibus database (accession number GSE322653).

The online version of this article contains a data supplement.

The publication costs of this article were defrayed in part by page charge payment. Therefore, and solely to indicate this fact, this article is hereby marked "advertisement" in accordance with 18 USC section 1734.

REFERENCES

- Swerdlow SH, Campo E, Pileri SA, et al. The 2016 revision of the World Health Organization classification of lymphoid neoplasms. *Blood*. 2016;127(20):2375-2390.
- Alizadeh AA, Eisen MB, Davis RE, et al. Distinct types of diffuse large B-cell lymphoma identified by gene expression profiling. *Nature*. 2000; 403(6769):503-511.
- Shaffer AL III, Young RM, Staudt LM. Pathogenesis of human B cell lymphomas. *Annu Rev Immunol*. 2012;30:565-610.
- Sehn LH, Salles G. Diffuse large B-cell lymphoma. *N Engl J Med*. 2021;384(9): 842-858.
- Chapuy B, Stewart C, Dunford AJ, et al. Molecular subtypes of diffuse large B cell lymphoma are associated with distinct pathogenic mechanisms and outcomes. *Nat Med*. 2018;24(5):679-690.
- Chapuy B, Wood TR, Stewart C, et al. DLBclass: a probabilistic molecular classifier to guide clinical investigation and practice in DLBCL. *Blood*. 2025;145(18):2041-2055.
- Schmitz R, Wright GW, Huang DW, et al. Genetics and pathogenesis of diffuse large B-cell lymphoma. *N Engl J Med*. 2018; 378(15):1396-1407.
- Wright GW, Huang DW, Phelan JD, et al. A probabilistic classification tool for genetic subtypes of diffuse large B cell lymphoma with therapeutic implications. *Cancer Cell*. 2020;37(4):551-568.e14.
- Lacy SE, Barrans SL, Beer PA, et al. Targeted sequencing in DLBCL, molecular subtypes, and outcomes: a Haematological Malignancy Research Network report. *Blood*. 2020; 135(20):1759-1771.
- Delbridge AR, Grabow S, Strasser A, Vaux DL. Thirty years of BCL-2: translating cell death discoveries into novel cancer therapies. *Nat Rev Cancer*. 2016;16(2): 99-109.
- Lee SH, Shin MS, Kim HS, et al. Somatic mutations of TRAIL-receptor 1 and TRAIL-receptor 2 genes in non-Hodgkin's lymphoma. *Oncogene*. 2001;20(3):399-403.

12. Testa U. TRAIL/TRAIL-R in hematologic malignancies. *J Cell Biochem.* 2010;110(1): 21-34.
13. Annibaldi A, Walczak H. Death receptors and their ligands in inflammatory disease and cancer. *Cold Spring Harb Perspect Biol.* 2020;12(9):a036384.
14. Strasser A, Jost PJ, Nagata S. The many roles of FAS receptor signaling in the immune system. *Immunity.* 2009;30(2):180-192.
15. Tummers B, Green DR. Caspase-8: regulating life and death. *Immunol Rev.* 2017;277(1):76-89.
16. Walczak H. Death receptor-ligand systems in cancer, cell death, and inflammation. *Cold Spring Harb Perspect Biol.* 2013;5(5): a008698.
17. Annibaldi A, Meier P. Checkpoints in TNF-induced cell death: implications in inflammation and cancer. *Trends Mol Med.* 2018;24(1):49-65.
18. Peltzer N, Annibaldi A. Cell death-related ubiquitin modifications in inflammatory syndromes: from mice to men. *Biomedicines.* 2022;10(6):1436.
19. Liccardi G, Annibaldi A. MLKL post-translational modifications: road signs to infection, inflammation and unknown destinations. *Cell Death Differ.* 2023;30(2): 269-278.
20. Scaffidi C, Fulda S, Srinivasan A, et al. Two CD95 (APO-1/Fas) signaling pathways. *Embo J.* 1998;17(6):1675-1687.
21. Irmeler M, Thome M, Hahne M, et al. Inhibition of death receptor signals by cellular FLIP. *Nature.* 1997;388(6638): 190-195.
22. Thome M, Schneider P, Hofmann K, et al. Viral FLICE-inhibitory proteins (FLIPs) prevent apoptosis induced by death receptors. *Nature.* 1997;386(6624):517-521.
23. Martinez Lagunas K, Savcigil DP, Zrilic M, et al. Cleavage of cFLIP restrains cell death during viral infection and tissue injury and favors tissue repair. *Sci Adv.* 2023;9(30): eadg2829.
24. Zhang H, Rosenberg S, Coffey FJ, et al. A role for cFLIP in B cell proliferation and stress MAPK regulation. *J Immunol.* 2009; 182(1):207-215.
25. Coffey F, Manser T. Expression of cellular FLIP by B cells is required for their participation in an immune response. *J Immunol.* 2010;184(9):4871-4879.
26. van Houdt IS, Muris JJ, Hesselink AT, et al. Expression of c-FLIP is primarily detected in diffuse large B-cell lymphoma and Hodgkin's lymphoma and correlates with lack of caspase 8 activation. *Histopathology.* 2007; 51(6):778-784.
27. Valente G, Manfroï F, Peracchio C, et al. cFLIP expression correlates with tumour progression and patient outcome in non-Hodgkin lymphomas of low grade of malignancy. *Br J Haematol.* 2006;132(5): 560-570.
28. Knittel G, Liedgens P, Korovkina D, et al. B-cell-specific conditional expression of Myd88p.L252P leads to the development of diffuse large B-cell lymphoma in mice. *Blood.* 2016;127(22):2732-2741.
29. Flumann R, Hansen J, Pelzer BW, et al. Distinct genetically-determined origins of Myd88/Bcl2-driven aggressive lymphoma rationalize targeted therapeutic intervention strategies. *Blood Cancer Discov.* 2023;4(1): 78-97.
30. Flumann R, Rehkämper T, Nieper P, et al. An autochthonous mouse model of Myd88- and BCL2-driven diffuse large B-cell lymphoma reveals actionable molecular vulnerabilities. *Blood Cancer Discov.* 2021;2(1):70-91.
31. Zhang N, He YW. An essential role for c-FLIP in the efficient development of mature T lymphocytes. *J Exp Med.* 2005;202(3): 395-404.
32. Brune V, Tiacchi E, Pfeil I, et al. Origin and pathogenesis of nodular lymphocyte-predominant Hodgkin lymphoma as revealed by global gene expression analysis. *J Exp Med.* 2008;205(10):2251-2268.
33. Jin H, Zhang C, Zwahlen M, et al. Systematic transcriptional analysis of human cell lines for gene expression landscape and tumor representation. *Nat Commun.* 2023;14(1): 5417.
34. Bertolo C, Roa S, Sagardoy A, et al. LITAF, a BCL6 target gene, regulates autophagy in mature B-cell lymphomas. *Br J Haematol.* 2013;162(5):621-630.
35. Painter D, Barrans S, Lacy S, et al. Cell-of-origin in diffuse large B-cell lymphoma: findings from the UK's population-based Haematological Malignancy Research Network. *Br J Haematol.* 2019;185(4): 781-784.
36. Haniuda K, Kitamura D. Induced germinal center B cell culture system. *Bio Protoc.* 2019;9(4):e3163.
37. Knittel G, Reinhardt HC. Genetic mouse models of lymphomas. *Methods Mol Biol.* 2025;2865:411-428.
38. Tabatabai A, Arora A, Höfmann S, et al. Mouse models of diffuse large B cell lymphoma. *Front Immunol.* 2023;14: 1313371.
39. Flumann R, Hansen J, Meinel J, et al. An inducible Cd79b mutation confers ibrutinib sensitivity in mouse models of Myd88-driven diffuse large B-cell lymphoma. *Blood Adv.* 2024;8(5):1063-1074.
40. Pindzola GM, Razzaghi R, Tavory RN, et al. Aberrant expansion of spontaneous splenic germinal centers induced by hallmark genetic lesions of aggressive lymphoma. *Blood.* 2022;140(10):1119-1131.
41. Henry CM, Martin SJ. Caspase-8 acts in a non-enzymatic role as a scaffold for assembly of a pro-inflammatory "FADDosome" complex upon TRAIL stimulation. *Mol Cell.* 2017;65(4):715-729.e5.
42. Davidovich P, Higgins CA, Najda Z, Longley DB, Martin SJ. cFLIP(L) acts as a suppressor of TRAIL- and Fas-initiated inflammation by inhibiting assembly of caspase-8/FADD/RIPK1 NF-κB-activating complexes. *Cell Rep.* 2023;42(12):113476.
43. Hartwig T, Montinaro A, von Karstedt S, et al. The TRAIL-induced cancer secretome promotes a tumor-supportive immune microenvironment via CCR2. *Mol Cell.* 2017; 65(4):730-742.e5.
44. Zhu H, Almasan A. Development of venetoclax for therapy of lymphoid malignancies. *Drug Des Devel Ther.* 2017;11: 685-694.
45. Schenk RL, Strasser A, Dewson G. BCL-2: long and winding path from discovery to therapeutic target. *Biochem Biophys Res Commun.* 2017;482(3):459-469.
46. Strasser A, Harris AW, Bath ML, Cory S. Novel primitive lymphoid tumours induced in transgenic mice by cooperation between myc and bcl-2. *Nature.* 1990;348(6299): 331-333.
47. Müschen M, Rajewsky K, Krönke M, Küppers R. The origin of CD95-gene mutations in B-cell lymphoma. *Trends Immunol.* 2002;23(2):75-80.

© 2026 American Society of Hematology. Published by Elsevier Inc. Licensed under Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International (CC BY-NC-ND 4.0), permitting only noncommercial, nonderivative use with attribution. All other rights reserved.